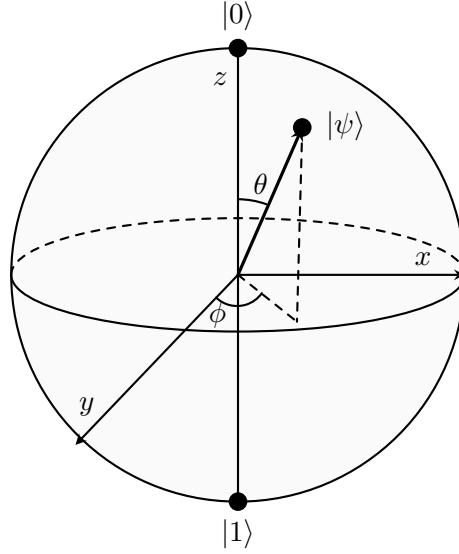


Physics 7501 Notes

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1 Bloch Sphere



The Bloch sphere is useful because it turns a two-level quantum state into geometry: measurement probabilities become projections, unitary evolution becomes rotation, and mixedness becomes distance from the center.

1.1 Visualizing a state as a point on the sphere

Choose $|0\rangle = |\uparrow\rangle$ and $|1\rangle = |\downarrow\rangle$. Any normalized pure qubit state, after removing an overall phase, can be written as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle$$

Its Bloch vector is

$$\mathbf{r} = (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta).$$

which is the unit radial vector in spherical coordinates. Here θ is the polar angle measured from $+z$, and ϕ is the azimuthal angle.

The half-angle in the ket makes it so that

- (i) $|0\rangle$ and $|1\rangle$ are the north and south poles
- (ii) $(|0\rangle \pm |1\rangle)/\sqrt{2}$ point along $\pm x$
- (iii) $(|0\rangle \pm i|1\rangle)/\sqrt{2}$ point along $\pm y$
- (iv) The Pauli expectations are equal to the cooresponding coordinate of the Bloch vector i.e. the projection onto the x, y, z axes

$$\langle X \rangle = \sin\theta \cos\phi \quad , \quad \langle Y \rangle = \sin\theta \sin\phi \quad , \quad \langle Z \rangle = \cos\theta$$

Orthogonal kets correspond to antipodal points on the sphere, not perpendicular Bloch vectors. Multiplying a ket by an overall phase leaves its point unchanged while changing the relative phase moves it.

Other useful facts:

(i) θ controls the measurement probabilities in the z -basis i.e.

$$P(Z = +1) = \cos^2(\theta/2) \quad , \quad P(Z = -1) = \sin^2(\theta/2)$$